Effective dimensionality $d_{\text{eff}}(x, \tau)$ from MGD's fitted θ_t

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Part I (Guth's d_{eff}, MGD equations, implementation) added 2026-10-15

Purpose

Guth et al. define a local effective dimensionality $d_{\text{eff}}(x, \tau)$ of a distribution's support, at a point x and noise scale τ, from the score of the variance-exploding (VE) noised distribution at x. This note derives d_{eff} directly in terms of MGD's own fitted quantities – θ_t, φ, $\nabla\varphi$ – so it can be computed as pure post-processing of an already-completed MGD run, with no new sampling/fitting. Implementation: `codes/effective_dimension.py`. Notebook (CSV loader for plots): `turbulence/effective_dimension_deff_final.ipynb`; the per-seed CSVs it reads are produced on the cluster by `deff_diagnostics/deff_latest_run_check.py`.

Headline conclusion (used in Section 12). The *denoiser form* (Eq. 19) is the right estimator to use in the paper. The *energy form* (Eq. 12) is strictly equivalent to it only when the model density $p_\theta(\cdot, \tau)$ satisfies the Fokker-Planck equation of the noising, which the moment-matched family produced by MGD does not. The measured energy–denoiser gap is a direct, quantitative read-out of this mismatch (Section 10), small at the trivial ends ($\tau \to 0, \infty$) and largest where the physics is doing the most work.

Reading order. Part I (Sections 1–4) is the self-contained reference. Part II is the earlier derivation and debugging record, kept for provenance. Where the two disagree, Part I holds.

Notation. Guth et al. call the noise variance t. Here it is τ, and $t \in [0, 1]$ is always MGD time, with $\alpha_t = \pi t/2$ and $\dot{\alpha} = \pi/2$. Signs follow the code: $p_\theta \propto \exp(+\theta^\top \varphi)$.

Part I: reference

1 A. Guth's effective dimensionality

Source: Guth, Kadkhodaie & Simoncelli, arXiv:2506.05310 (copy in `turbulence/guth_et_al_paper.pdf`), Sec. 3.3 and App. B.3.

Noise a clean point $x \in \mathbb{R}^d$ as $y = x + \sqrt{\tau}\,\xi$, with $\xi \sim \mathcal{N}(0, I_d)$. Let $q_\tau = p * \mathcal{N}(0, \tau I)$ be the noisy density and $U = -\log q_\tau$ its energy. The optimal denoiser is the posterior mean, given by Tweedie's formula: $\mathbb{E}[x \mid y] = y + \tau \nabla \log q_\tau(y)$. The two definitions (their Eqs. 11–12) are

$$d_{\text{eff}}^{\text{den}}(x, \tau) = \frac{1}{\tau}\,\mathbb{E}_\xi \big\| x - \mathbb{E}[x \mid y] \big\|^2, \qquad d_{\text{eff}}^{\text{en}}(x, \tau) = d - 2\tau\,\partial_\tau\,\mathbb{E}_\xi \big[U(x + \sqrt{\tau}\,\xi, \tau) \big]. \tag{1}$$

Meaning. Suppose that at scale τ the data near x fill a k-dimensional subspace. The denoiser projects onto it and keeps the k noise components along it, so the error is τk and $d_{\text{eff}} = k$. $d_{\text{eff}} \to 0$ for $\tau \to \infty$, and $d_{\text{eff}} \to d$ for $\tau \to 0$ when the density is smooth and positive at x.

Equivalence and its condition (App. B.3). Stein's lemma turns the first definition into $d + \tau\,\mathbb{E}_\xi[\|\nabla U\|^2 - 2\Delta U]$. This holds for *any* smooth U. Differentiating $\mathbb{E}_\xi[U]$ gives the same expression only if U solves the heat equation written for the energy, $\partial_\tau U = \frac{1}{2}\Delta U - \frac{1}{2}\|\nabla U\|^2$. In general

$$d_{\text{eff}}^{\text{en}} - d_{\text{eff}}^{\text{den}} = -2\tau\,\mathbb{E}_\xi[R], \qquad R = \partial_\tau U - \tfrac{1}{2}\Delta U + \tfrac{1}{2}\|\nabla U\|^2. \tag{2}$$

MMSE property. The posterior mean minimises the squared error averaged over $x \sim p$. Any other denoiser D therefore gives $\mathbb{E}_x \frac{1}{\tau}\mathbb{E}_\xi\|x - D(y)\|^2 \;\geq\; \mathbb{E}_x d_{\text{eff}}(x,\tau)$. There is no such ordering pointwise in x.

2 B. What is needed from MGD

Interpolant and moments. $I_t = \cos\alpha_t\, x_0 + \sin\alpha_t\, x_1$, with $x_0 \sim \mathcal{N}(0, I)$ and $x_1 \sim$ data. The moments are $m_t = \mathbb{E}[\varphi(I_t)]$ and $\dot m_t = dm_t/dt$. The Gram matrix on walkers X is $G(X) = \overline{\nabla\varphi\,\nabla\varphi^\top}$. An overbar denotes the mean over the N walkers.

One SDE step (`iteration_step_projection`), with step h and noise level σ:

$$G(x_k)\,\eta_k = \dot m_{t_k}, \qquad y_k = x_k + h\,\nabla\varphi^\top\eta_k + \sqrt{2h}\,\sigma\,\xi_k, \tag{3}$$

$$G(y_k)\,\theta_k = \frac{m_{t_{k+1}} - \bar\varphi(y_k)}{h\sigma^2}, \qquad x_{k+1} = y_k + h\sigma^2\,\nabla\varphi^\top\theta_k. \tag{4}$$

The walker drift has two parts. The first, $\nabla\varphi^\top\eta_k$, is a transport velocity that reproduces $\dot m_t$. The second, $\sigma^2\nabla\varphi^\top\theta_k$, together with the noise, is a Langevin step for the density $e^{\theta_k^\top\varphi}$.

What θ_k estimates. Apply Itô's formula to $\bar\varphi$ over the predictor step, and assume the previous corrector hit its target, $\bar\varphi(x_k) = m_{t_k}$: $\bar\varphi(y_k) = m_{t_k} + h\dot m_{t_k} + h\sigma^2\,\overline{\Delta\varphi} + \varepsilon_k$. Here ε_k is the walker average of $\sqrt{2h}\sigma\,\nabla\varphi\cdot\xi_k$. Inserted into Eq. 4:

$$G\,\theta_k = -\overline{\Delta\varphi} \;-\; \frac{\varepsilon_k}{h\sigma^2} \;+\; O(h). \tag{5}$$

Write ρ_t for the walker law. $G\theta = -\overline{\Delta\varphi}$ are the normal equations of Hyvärinen score matching, $\min_\theta \mathbb{E}_{\rho_t}\|\theta^\top\nabla\varphi - \nabla\log\rho_t\|^2$. So $\theta_k^\top\nabla\varphi$ is a noisy estimate of the projection of the walkers' score onto $\text{span}\{\nabla\varphi_j\}$. The noise has size $\|\varepsilon_k\|/(h\sigma^2) \sim \|\nabla\varphi\|/(\sigma\sqrt{hN})$. MGD assumes $\rho_t = p_{\theta_t} \propto e^{\theta_t^\top\varphi}$, the maximum-entropy law with moments m_t. Under that assumption the projection is exact:

$$\nabla\log p_{\theta_t} = \theta_t^\top\nabla\varphi. \tag{6}$$

Time-regularised Θ_{reg} (`forward_regularised`, `reg_mode='moment'`). It minimises

$$\int_0^1 \Big[\tfrac{1}{2}\Theta^\top M_t\Theta - b_t^\top\Theta\Big]dt + \lambda\int_0^1\Big[\tfrac{1}{2}\dot\Theta^\top\Sigma_t\dot\Theta - \dot m_t^\top\dot\Theta\Big]dt, \tag{7}$$

where $M_k = G(y_k)$ and b_k is the right-hand side of Eq. 4. The first term alone gives back θ_k, i.e. the per-step score matching. In the second term, $\Sigma_t = \text{Cov}(\varphi)$ on the walkers. For an exponential family $\frac{d}{dt}\mathbb{E}_{p_\Theta}[\varphi] = \text{Cov}(\varphi)\dot\Theta$, so the second term's normal equation $\Sigma\dot\Theta = \dot m$ asks the family to follow the moments in time. This is the analogue of Guth's time-score loss. λ is chosen after the run.

Normaliser (energy form only). $H(p_\theta) = \log Z - \theta^\top m$, and $dH/dt = -\theta_t^\top \dot{m}_t$ while the moments are followed, so

$$\log Z_t = \tfrac{d}{2}\log(2\pi e) - \int_0^t \theta_u^\top \dot{m}_u \, du + \theta_t^\top m_t. \tag{8}$$

The integrand is saved per step as `dH_t_bound`. `h_bound_at` integrates it with θ_k. `h_from_theta_reg` integrates it with Θ_{reg}.

3 C. d_{eff} written in MGD variables, and the code

The link. Take $x_1 = x$, the clean point. Then $I_t = \sin\alpha_t\, x + \cos\alpha_t\, \xi = \sin\alpha_t\,(x + \cot\alpha_t\, \xi)$. This is Guth's noisy observation, multiplied by $\sin\alpha_t$, at

$$\tau = \cot^2\alpha_t \qquad (\tau = \infty \text{ at } t = 0,\ \tau = 1 \text{ at } t = \tfrac{1}{2},\ \tau = 0 \text{ at } t = 1). \tag{9}$$

Rewriting Tweedie in these variables (exact for the law p_t of I_t) gives $\mathbb{E}[x \mid I_t] = \big(I_t + \cos^2\alpha_t \nabla \log p_t(I_t)\big)/\sin\alpha_t$.

Denoiser form. Then $x - \mathbb{E}[x \mid I_t] = -\cot\alpha_t\,(\xi + \cos\alpha_t \nabla \log p_t(I_t))$. Divide by $\tau = \cot^2\alpha_t$ and use Eq. 6 with $\theta_t = \Theta_{\text{reg}}(t)$:

$$\boxed{d_{\text{eff}}^{\text{den}}(x,t) = \mathbb{E}_\xi \big\| \, \xi + \cos\alpha_t\, \theta_t^\top \nabla\varphi\big(\sin\alpha_t\, x + \cos\alpha_t\, \xi\big)\big\|^2} \tag{10}$$

Check. For Gaussian data $\mathcal{N}(0, \sigma_{\text{d}}^2 I)$, $p_t = \mathcal{N}(0, v_t I)$ with $v_t = \cos^2\alpha_t + \sigma_{\text{d}}^2 \sin^2\alpha_t$ and $\nabla \log p_t(I) = -I/v_t$. Then Eq. 10 gives $\sum_i (\tau x_i^2 + \sigma_{\text{d}}^4)/(\sigma_{\text{d}}^2 + \tau)^2$, which is Eq. 25.

Energy form. $U = -\log q_\tau(y) = -\log p_t(I_t) - d\log\sin\alpha_t$ and $2\tau\partial_\tau = -\frac{\sin\alpha_t\cos\alpha_t}{\dot{\alpha}}\,\partial_t$. Hence

$$d_{\text{eff}}^{\text{en}}(x,t) = d\sin^2\alpha_t - \frac{\sin\alpha_t\cos\alpha_t}{\dot{\alpha}}\,\frac{d}{dt}\,\mathbb{E}_\xi\Big[\theta_t^\top \varphi\big(\sin\alpha_t\, x + \cos\alpha_t\, \xi\big) - \log Z_t\Big], \tag{11}$$

with ξ held fixed inside the derivative.

Code (`codes/effective_dimension.py`).

- `denoiser_d_eff`: Eq. 10 at each requested node of t_{reg}, with n_{mc} fresh ξ per node. $\theta_t^\top \nabla\varphi$ is computed by `grad_phi_projected`, in float64 where the potential accepts it. It is written as $\xi + \sqrt{\tau}\sin\alpha_t\, \theta_t^\top \nabla\varphi$, which is the same thing since $\sqrt{\tau}\sin\alpha_t = \cos\alpha_t$.
- `energy_d_eff`: $d - \partial_{\frac{1}{2}\log\tau}$ of the Monte Carlo mean of $-\theta_t^\top\varphi + \log Z_t - d\log\sin\alpha_t$, i.e. Eq. 11 with the derivative taken numerically in $\frac{1}{2}\log\tau$ instead of t (`local_poly_derivative`). The same ξ is used at all nodes (`common_noise=True`). $\log Z_t$ comes from Eq. 8.
- Driver `deff_diagnostics/deff_latest_run_check.py`:
  - loads `fitted_potentials/` (without them the fitted r, and so every index into θ, is wrong);
  - loads $\Theta_{\text{reg}}, t_{\text{reg}}$, and takes $m_t =$ `barphi_e` at the t_{reg} nodes;
  - evaluates at training points $x \in x_1$ split typical/rare, on nodes uniform in $\log\tau$;
  - writes per-seed CSVs, which `turbulence/effective_dimension_deff_final.ipynb` plots.

4 D. Subtleties of the translation, and the choice made for each

1. **Which density d_{eff} describes.** At time t there are three laws: the true law of I_t (noised data), the walker law ρ_t, and the model p_{θ_t}. Guth's $\mathbb{E}[x|y]$ uses the first. Only the third has a score available in closed form. *Choice:* use the model score, but draw the noise exactly as Guth does: $I_t = \sin\alpha_t x + \cos\alpha_t \xi$ with x clean. Eq. 10 is then the normalised error of the model's denoiser $D_t(I) = (I + \cos^2\alpha_t\, \theta_t^\top \nabla\varphi(I))/\sin\alpha_t$, measured on true noisy observations of x. This is a property of the *model*, not of the data.

2. **The model denoiser is not a posterior mean.** The p_{θ_t} are separate fits at each t. They are not one prior diffused by the heat equation, so D_t is not $\mathbb{E}[x|I_t]$ for any prior. This is the situation of Guth's Eq. 16 (Tweedie with a learned score). Three consequences follow.

   - The denoiser form stays well defined: it is the error of a denoiser that exists.
   - The energy form differs from it by Eq. 2, because $R \neq 0$.
   - Averaged over data points, $\mathbb{E}_x d_{\text{eff}}^{\text{den}} \geq \mathbb{E}_x d_{\text{eff}}^{\text{true}}$ (MMSE property). The model value is biased upward on average, with no ordering pointwise.

   Choice: report $d_{\text{eff}}^{\text{den}}$, and keep $d_{\text{eff}}^{\text{en}}$ only as a read-out of R.

3. **"Moment-matched, not score-matched" is too coarse.** By Eq. 5, θ_k *is* a score-matching estimate, but of the walker law, inside $\text{span}\{\nabla\varphi_j\}$. The data enter only through m_t. The model score therefore agrees with the data score only to the extent that the moments pin it down. This replaces the corresponding sentence in Part II.

4. **Only θ enters, not η.** The transport part $\nabla\varphi^\top \eta_k$ of the drift is a velocity for the moment path, not a score. Tweedie needs the score, which is the Langevin part divided by σ^2. *Choice:* use θ normalised by $h\sigma^2$, as the code stores it. Θ_{reg} is built from b_k with the same normalisation.

5. **Θ_{reg}, not θ_k.** The step noise in Eq. 5 grows as $h \to 0$. $d_{\text{eff}}^{\text{den}}$ is quadratic in θ, so zero-mean noise δ in θ does not average out. It adds $\cos^2\alpha_t\, \mathbb{E}\|\delta^\top \nabla\varphi\|^2 > 0$: an upward bias. *Choice:* use Θ_{reg}. The price is that results depend on λ. The λ-term is also the only ingredient that pushes the family toward time consistency, so it directly affects R and hence the energy–denoiser gap.

6. **VP interpolant vs VE noise.** The rescaling by $\sin\alpha_t$ is exact, and it leaves a factor $\cos\alpha_t$ in front of the score (Eq. 10). The points where $\nabla\varphi$ is evaluated, $\sin\alpha_t x + \cos\alpha_t \xi$, are interpolant samples with $x_1 = x$, so for x in the data set they lie where the fit had data. *Choice:* no change of variables in the code. It works directly on I_t.

7. **Energy form: what must be held fixed, and $\log Z$.**

   - The expectation is over ξ at fixed x, so the t-derivative is taken path by path at fixed ξ. `common_noise=True` is therefore the estimator of the right quantity, not only a variance trick.
   - Only $d\log Z_t/dt$ matters. With `h_bound_at` that derivative is $\dot\Theta^\top m + (\Theta_{\text{reg}} - \theta_k)^\top \dot m$, i.e. it mixes the two θ paths. `h_from_theta_reg` removes the mixing but is not covered by MGD's entropy-bound proof.
   - The code differentiates in $\frac{1}{2}\log\tau$, which is very unevenly spaced on t_{reg} (Part II). Eq. 11 differentiates in t, a possible improvement that is *not* implemented.

8. **Evaluation points.** The driver uses points of the training set x_1, whereas Guth et al. use held-out images. With a few hundred parameters the difference should be small, but it has not been checked.

Correspondence.

Guth et al.	MGD	code
noise variance τ	$\cot^2 \alpha_t$	`tau_of_t`
$y = x + \sqrt{\tau}\xi$	$I_t / \sin \alpha_t$, with $I_t = \sin \alpha_t x + \cos \alpha_t \xi$	inline
learned score	$\Theta_{\text{reg}}(t)^\top \nabla \varphi(I_t)$ (in I_t units)	`grad_phi_projected`
denoising loss	per-step score matching, first term of Eq. 7	`compute_theta`
time-score loss	λ-term of Eq. 7	`_solve_regularised_thomas`
Eq. 11	Eq. 10	`denoiser_d_eff`
Eq. 12	Eq. 11	`energy_d_eff`

Part II: derivation and debugging record

5 Guth's d_{eff}: recap

Let q_τ be the law of $y = x + \sqrt{\tau}\,\xi$, $\xi \sim \mathcal{N}(0, I_d)$, x drawn from the data. Two equivalent definitions:

$$\text{energy form:} \qquad d_{\text{eff}}(x, \tau) = d - \partial_s \, \mathbb{E}_y\big[U(y, \tau) \mid x\big], \qquad s = \tfrac{1}{2} \log \tau, \tag{12}$$

$$\text{denoiser form:} \qquad d_{\text{eff}}(x, \tau) = \frac{1}{\tau} \, \mathbb{E}_y\big[\|x - \mathbb{E}[x \mid y]\|^2 \mid x\big], \tag{13}$$

where $U(\cdot, \tau) = -\log q_\tau(\cdot)$ and, by Tweedie's formula, $\mathbb{E}[x \mid y] = y + \tau \, \nabla_y \log q_\tau(y)$. Both reduce to d for an isotropic Gaussian and to (less than) d when the data lies near a lower-dimensional manifold, at scales τ finer than the manifold's local curvature.

6 MGD's schedule and sign convention

MGD's Cos stochastic interpolant (`sde_routines.py`, `compute_ interpolant`, `interpolant='Cos'` – the only schedule `forward_regularised` supports) is

$$I_t = \cos(\alpha_t)\, x_0 + \sin(\alpha_t)\, x_1, \qquad \alpha_t = \tfrac{\pi t}{2}, \qquad x_0 \sim \mathcal{N}(0, I_d), \; x_1 = x \text{ (data)}. \tag{14}$$

Sign convention. This codebase fits θ_t for $p_\theta(x) \propto \exp(+\theta^\top \varphi(x))$ – the *opposite* of the MGD paper's $\exp(-\theta^\top \varphi(x))$. This is audited directly against the code in `codes/utils_entropy.py` ("Sign convention audit"): the corrector step in `iteration_step_projection` is $x_{k+1} = y_k + \nabla\varphi^\top \theta_{\text{raw}}$ (a *plus*), giving a continuous-time drift $\sigma^2 \nabla\varphi^\top \theta_t$ that must equal $\sigma^2 \nabla \log p_t$ for the transport to preserve p_t, so

$$\nabla_x \log p_t(x) \;=\; +\theta_t^\top \nabla \varphi(x). \tag{15}$$

Independently corroborated by $dH_k = -\theta_k^\top \dot{m}_t$ (the entropy increment already computed in `iteration_step_projection`) and by $\log Z_{\theta_1} \gtrsim H_{\text{bound}} + \theta_1^\top m_1$ (MGD Eq. 8/30 as implemented in `utils_entropy.log_Z_bound`). Every formula below is re-derived directly in this (+) convention – not by substituting $\theta_{\text{paper}} = -\theta_{\text{code}}$ into the paper's stated formulas.

7 The VE rescaling

Define $y_t := I_t / \sin(\alpha_t) = x_1 + \cot(\alpha_t)\, x_0$. Conditional on $x_1 = x$,

$$y_t \mid x \;\sim\; \mathcal{N}\big(x,\; \tau(t)\, I_d\big), \qquad \tau(t) = \cot^2(\alpha_t), \tag{16}$$

monotonically decreasing from $\tau(0) = \infty$ to $\tau(1) = 0$. So MGD's Cos schedule already sweeps every VE noise scale, and t_{reg} (the grid θ_t is fit on) IS a τ-grid via Eq. 16.

Write q_τ (Guth's VE marginal, density in y) in terms of p_t (MGD's marginal, density in I_t) by the change of variables $I_t = \sin(\alpha_t)\, y$:

$$q_\tau(y) = \sin^d(\alpha_t)\; p_t\big(\sin(\alpha_t)\, y\big). \tag{17}$$

Differentiating and using Eq. 15 (evaluated at $I_t = \sin(\alpha_t) y$):

$$\nabla_y \log q_\tau(y) \;=\; \sin(\alpha_t)\, \nabla_x \log p_t(\sin(\alpha_t) y) \;=\; +\sin(\alpha_t)\, \theta_t^\top \nabla\varphi\big(\sin(\alpha_t)\, y\big). \tag{18}$$

The extra $\sin(\alpha_t)$ factor is the chain-rule term the derivation asked to double-check – confirmed, with the (+) sign from Eq. 15 (a paper-convention derivation would carry a minus here).

Practical note. $\sin(\alpha_t)\, y = \sin(\alpha_t)\, x + \cos(\alpha_t)\, \xi$ for $y = x + \sqrt{\tau}\, \xi$ (since $\sqrt{\tau} = \cot \alpha_t$) – literally `compute_interpolant`'s own Cos schedule with x as a single-point data endpoint and ξ as fresh noise. So every $\varphi/\nabla\varphi$ evaluation below stays inside the numerical range the fit itself already explores; nothing is evaluated off-distribution in an uncontrolled way.

8 Denoiser form

Substituting Eq. 18 into Tweedie's formula, $x - \mathbb{E}[x \mid y] = -\sqrt{\tau}\, \xi - \tau\, \nabla_y \log q_\tau(y)$, so the $1/\tau$ in Eq. 13 cancels analytically:

$$\boxed{\; d_{\text{eff}}(x, \tau) \;=\; \mathbb{E}_\xi\Big[\; \big\| \,\xi + \sqrt{\tau}\, \sin(\alpha_t)\, \theta_t^\top \nabla\varphi\big(\sin(\alpha_t)\, y\big) \,\big\|^2 \;\Big], \qquad y = x + \sqrt{\tau}\, \xi. \;} \tag{19}$$

No $\log Z_t$, no m_t, no normalisation constant – only θ_t and $\nabla\varphi$. This is the primary estimator (`effective_dimension.denoiser_d_eff`).

9 Energy form

From $U(y, \tau) = -\log q_\tau(y)$,

$$U(y, \tau) \;=\; -\theta_t^\top \varphi\big(\sin(\alpha_t) y\big) \;+\; \log Z_t \;-\; d \log \sin(\alpha_t). \tag{20}$$

The last term is analytic in t; the first two are not, so Eq. 12's ∂_s is taken numerically, by finite-differencing $\mathbb{E}_y[U(y, \tau) \mid x]$ directly across the t_{reg} grid (converted to $s = \frac{1}{2} \log \tau(t)$), rather than splitting analytic/Monte-Carlo pieces (`effective_dimension.energy_d_eff`).

$\log Z_t$ in the pipeline. Our pipeline supplies $\log Z_t$ via MGD's Prop. 4.3/Eq. 30 construction,

$$\log Z_t \;=\; H_{\text{bound}}(t) + \theta_t^\top m_t, \qquad H_{\text{bound}}(t) = H(p_0) + \int_0^t dH_s\, ds, \tag{21}$$

generalised from the paper's stated $t = 1$ endpoint to any t (the integral is just truncated earlier). $H_{\text{bound}}(t)$ is built by trapezoid-integrating the already-saved dH_t^{bound} (`effective_dimension.h_bound_at`); m_t is either the saved `aux_moments` (the run's own realised target moments) or, if not pulled from the cluster yet, recomputed from the data and schedule, $m_t = \mathbb{E}_Z[\varphi(\cos\alpha_t\, Z + \sin\alpha_t\, x_1)]$ – independent of any saved particle trajectory, so no SDE rerun either way (`effective_dimension.recompute_m_t`). An alternative, θ_t-only construction $\log Z_t = -H(p_0) - \int_0^t \theta_s^\top \dot{m}_s\, ds$ is also implemented (`effective_dimension.h_from_theta_reg`); the two choices differ by less than 0.3 nats on the production runs we have measured (an x-independent shift $-d(\delta \log Z)/ds$ of the energy-form curve), so neither is the explanation for the much larger disagreement we discuss next.

10 The energy–denoiser gap: a direct read-out of model residual

Why Eqs. 12 and 13 agree in Guth's setting. Their equivalence is proved (their Appendix B.3) by one integration-by-parts step that uses the Fokker–Planck equation of the noising:

$$\partial_\tau\, q_\tau(y) \;=\; \tfrac{1}{2}\Delta_y\, q_\tau(y). \tag{22}$$

For a density q_τ obtained by heat-diffusing a single underlying q_0, Eq. 22 holds exactly and the two definitions coincide analytically.

Why they do *not* agree for MGD. Our pipeline does not produce a density that solves Eq. 22 along τ. At each τ we solve a dual problem that fixes a few averages. In interpolant coordinates $z = \sin\alpha_t\, y$:

$$p_{\theta_t}(z) \;\propto\; \exp\big(+\theta_t^\top \varphi(z)\big), \qquad \mathbb{E}_{p_{\theta_t}}[\varphi] \;=\; m_t, \qquad q_\tau(y) = \sin^d\!\alpha_t\; p_{\theta_t}(\sin\alpha_t\, y). \tag{23}$$

That is, the family $\{p_{\theta_t}\}$ is *moment-matched* to the true noisy data, not *score-matched*. The two conditions are related but strictly inequivalent: moment matching constrains a finite number of integrals $\int \varphi_j\, p_\theta$; score matching asks that $\nabla_y \log p_\theta$ agree with $\nabla_y \log q_\tau$ pointwise in y, which is much stronger. MGD's θ_t is a score-matching estimate, but of the *walker* law and inside $\operatorname{span}\{\nabla\varphi_j\}$; the data enter only through m_t (Section 4, item 3). So the model score matches the data score only as far as the moments constrain it.

The residual is the Hamilton–Jacobi residual of Part I, $R = \partial_\tau U - \frac{1}{2}\Delta_y U + \frac{1}{2}\|\nabla_y U\|^2$ (Eq. 2). It enters the gap exactly as $d_{\text{eff}}^{\text{en}} - d_{\text{eff}}^{\text{den}} = -2\tau\, \mathbb{E}_\xi[R]$ (Eq. 2). Take U from Eq. 20, i.e. $U = -\theta_t^\top\varphi(z) + \log Z_t - d\log\sin\alpha_t$ with $z = \sin\alpha_t\, y$. Use $\partial_\tau|_y = (dt/d\tau)\,\partial_t|_y$, $dt/d\tau = -\sin^3\!\alpha_t/(2\dot\alpha\cos\alpha_t)$, $\dot\alpha = \pi/2$, $\nabla_y = \sin\alpha_t \nabla_z$ and $\partial_t z|_y = \dot\alpha\cot\alpha_t\, z$. Write $\dot\theta_t = d\theta_t/dt$ and $\delta_t := \partial_t \log Z_t - \dot\theta_t^\top m_t$ ($\delta_t = 0$ when $\log Z_t$ is consistent with the moments). Then

$$\begin{aligned} R(y,\tau) \;=\; & \frac{\sin^3\alpha_t}{2\dot\alpha\cos\alpha_t}\Big[\; \underbrace{\dot\theta_t^\top\big(\varphi(z) - m_t\big)}_{\text{(i) rate of change of }\theta} \;-\; \underbrace{\delta_t}_{\text{(v) }\log Z\text{ drift}} \;\Big] \\ & + \tfrac{1}{2}\sin^2\alpha_t\Big[\; \underbrace{\theta_t^\top \Delta_z\varphi(z)}_{\text{(ii) basis curvature}} \;+\; \underbrace{\big\|\theta_t^\top\nabla_z\varphi(z)\big\|^2}_{\text{(iii) score magnitude}} \;+\; \underbrace{z\cdot\theta_t^\top\nabla_z\varphi(z) + d}_{\text{(iv) rescaling }I_t=\sin\alpha_t\, y} \;\Big]. \end{aligned} \tag{24}$$

Term (i) measures whether the Lagrange multiplier moves in step with the moments at the point z. Term (ii) is the Laplacian of the basis functions weighted by their multipliers, a curvature that moment matching never constrains. Term (iii) is the squared score. Term (iv) comes from the $\sin\alpha_t$ rescaling between MGD's I_t and Guth's y, through both the y-dependence of z and the Jacobian

$-d \log \sin \alpha_t$. Term (v) is nonzero when $\log Z_t$ is not built from the same θ path (e.g. `h_bound_at` with per-step θ_k while $\theta_t = \Theta_{\text{reg}}$; then $\delta_t = (\Theta_{\text{reg}} - \theta_k)^\top \dot{m}_t$). *Check.* Take $\varphi(z) = -\frac{1}{2}\|z\|^2$, $\theta_t = 1/v_t$ and $v_t = \cos^2 \alpha_t + \sigma_{\text{data}}^2 \sin^2 \alpha_t$ (the exact Gaussian case of Section 11). Then the first bracket gives $\frac{\sin^4 \alpha_t (\sigma_{\text{data}}^2 - 1)}{2 v_t^2}(\|z\|^2 - d v_t)$ and the second gives its negative, so $R \equiv 0$. For a true heat-equation solution the terms always cancel. For the moment-matched family they generically do not.

What is measured on the production runs. The energy and denoiser forms (Eqs. 12 and 13) use the *same* θ_t, *same* φ and *same* evaluation grid; they can be compared point-for-point. On the final turbulence runs (d=256, λ=5×10^{-4}, 5 seeds, with the x-dependent Monte-Carlo noise cancelled by common random numbers across τ-nodes), the mean gap $d_{\text{eff}}^{\text{energy}} - d_{\text{eff}}^{\text{denoiser}}$ is

- at most a few units at the extremes $\tau \to 0$ and $\tau \to \infty$, where both forms either saturate at d or vanish;
- of order -50 to -70 in the mid-τ range $\tau \in [10^{-4}, 10^{-2}]$, with per-point standard deviation ~ 200;
- reproducible across arms (same signal with λ=10^{-3} constrained L_6<0 *and* λ=5×10^{-4} unconstrained),

exactly tracking the location of the physical burst scale in the data (the bump in $d_{\text{eff}}^{\text{denoiser}}$ sits at the same τ). The two $\log Z_t$ constructions differ by ≤ 0.3 nats in $\log Z_t$ itself. They shift $d_{\text{eff}}^{\text{energy}}$ by $-\partial_s$ of that difference (term (v) above), which is negligible next to a gap of tens of units of d_{eff}. The remainder is the integrated residual $-2\tau\, \mathbb{E}_\xi[R]$ of Eq. 24, and it is *structurally present* in any moment-matched family: it is the quantitative cost of substituting a dual on $\theta \cdot m$ for the stronger Fisher-divergence objective a score network would minimise.

Physical reading. The gap is largest exactly where our denoiser-form d_{eff} shows a distinct bump above its monotone trend – at the burst scale $\tau \sim 10^{-3}$, in the mid-τ window where the noising transport actively exchanges information with the data manifold. There, several moment constraints are active simultaneously (second-order moments, scattering coefficients across adjacent scales, the L_6 intermittency term), the Lagrange multipliers change fastest, and Eq. 24's terms (i) and (iii) peak together. At both extremes the family trivialises: $p_\theta(\cdot, \tau)$ is nearly Gaussian in y-coordinates at large τ (one moment active) and nearly a delta at the data at small τ, and in both limits the contributions in Eq. 24 cancel to within a few units. The gap is therefore not a numerical artefact; it is the finger-print of how moment-matched p_θ differs from the exact noisy marginal, with amplitude controlled by the physics of the transport.

11 Validation against a closed-form Gaussian test

To separate genuine implementation bugs from the expected bound bias above, both estimators were checked against an exact, independently-derived ground truth – no SDE, no fitting, no θ_t from any real run. For $x \sim \mathcal{N}(0, \Sigma)$, $y = x + \sqrt{\tau}\,\xi$, $q_\tau = \mathcal{N}(0, \Sigma + \tau I)$ exactly, so Tweedie's formula gives (diagonal basis, $\Sigma = \text{diag}(\sigma_i^2)$):

$$d_{\text{eff}}(x, \tau) \;=\; \sum_i \frac{\tau x_i^2 + \sigma_i^4}{(\sigma_i^2 + \tau)^2}, \tag{25}$$

verified against brute-force $(\Sigma + \tau I)^{-1}$ linear algebra (agrees to $\sim 10^{-16}$) and its $\tau \to 0$ ($\to d$) and isotropic-expectation ($\to d\sigma^2/(\sigma^2 + \tau)$) limits. For *this* synthetic data MGD's own p_t is also exactly

Gaussian, so $\theta_t = -d/(2v(t))$ and $\log Z_t = \frac{d}{2}\log(2\pi v(t))$, $v(t) = \cos^2\alpha_t + \sin^2\alpha_t\,\sigma^2_{\text{data}}$, are exact too (single-coefficient potential $\varphi(x) = \overline{x^2}$) – feeding these directly into the *unmodified* `denoiser_d_eff`/`energy_d_eff` isolates implementation bugs with zero fitting noise and zero bound bias.

Using the real production grid (d=256, t_{reg} reconstructed from `nt=40000`, `n_subsample=200`): **denoiser_d_eff matched the oracle to $< 0.5\%$ relative error everywhere** (10 decades of τ, $|\theta|$ up to 512) – its implementation is correct. `energy_d_eff`, despite the exact inputs, showed errors up to 1320%, concentrated at the *coarsest* end of the grid.

11.1 Root cause: $s = \frac{1}{2}\log\tau$ is highly non-uniform, and it was differentiated naively

$\tau(t) = \cot^2(\alpha_t)$ diverges (logarithmically, in s) at both $t \to 0$ and $t \to 1$, so $s(t)$ is intrinsically badly stretched near the ends of the range *regardless of the t-schedule* – confirmed directly: a perfectly *uniform*-in-t grid has an even worse max/min local-Δs ratio (340×) than the actual Cos-schedule production grid (67×, since bunching points near $t = 1$ partially compensates). Differentiating $\mathbb{E}_y[U]$ across this grid with a raw finite difference (`numpy.gradient`, which defaults to `edge_order=1`) has two compounding failure modes, both confirmed in isolation:

(A) **Boundary truncation bias.** `edge_order=1` is only first-order accurate at the array's first/last nodes – on a noiseless synthetic quadratic in s, this alone produces a 6.4% error at the boundary (exact at every interior node).

(B) **Monte-Carlo noise amplification** (the dominant effect): $U(t)$ is itself MC-estimated, and dividing its noise by a small local Δs amplifies it. An n_{mc} sweep at the worst node showed the error shrinking monotonically as n_{mc} grew (0.77 → 0.29 → 0.23 over 4k→16k→32k), confirming noise, not just truncation.

The natural first fix, `edge_order=2`, removes (A) exactly (boundary error drops to machine precision on the noiseless check) but **does not help, and can make the realistic case worse** (max relative error 17.9 vs. 10.4) – its wider stencil has larger coefficients under uneven spacing, amplifying (B) more than it fixes (A).

11.2 Fix: a local polynomial derivative

`energy_d_eff` now differentiates $\mathbb{E}_y[U]$ via `effective_dimension.local_poly_derivative`: at each node, a weighted-least-squares quadratic is fit through a window of $2h + 1$ *neighbouring indices* (shifted, not shrunk, at the array edges, so every node – including the boundary – gets a full window) and differentiated analytically. This fixes (A) exactly (machine precision on the noiseless check, any window size) and reduces (B) by averaging over several points rather than two. Validated on the Gaussian test ($h = 5$, the shipped default): overall median relative error dropped $\sim 6\times$ (1.4% → 0.22%), and the fraction of the grid accurate to $< 1\%$ roughly doubled (39% → 70% of nodes, with the "becomes reliable" point moving from $t \approx 0.80$ down to $t \approx 0.41$). The few nodes at the very coarsest end of the grid remain the least reliable part of an energy-form curve on principle – no local model, however fit, is well-conditioned there – and should be read with that in mind; the denoiser form has no such issue, since it involves no differentiation step at all.

12 Which form to use in the paper

Recommendation. Use the denoiser form (Eq. 13, implemented in `denoiser_d_eff`) as the primary estimator. The energy form (Eq. 12) belongs in an appendix or methods section as a

diagnostic, not in a main figure. Three reasons:

1. *Unbiased for the quantity we care about.* What we want to report is a well-defined property of our *learned* model p_θ: how much of x remains after one scale of noising, averaged over noise draws. That is literally Eq. 13, and it uses only the model's score $+\sin\alpha_t\,\theta_t^\top\nabla\varphi$ via Tweedie's formula – no assumption about whether $p_\theta(\cdot,\tau)$ is the exact noisy marginal of anything. The energy form (Eq. 12) silently assumes p_θ satisfies Eq. 22, which MGD's moment-matched family does not (Section 10). Reporting the energy form means reporting a quantity biased by the model's FPE residual, with a mandatory footnote explaining that the bias is physical rather than numerical.

2. *Closer to the MGD construction.* MGD is defined by moment matching and generates samples by running an SDE whose drift is the model score. The denoiser form reads that same score through Tweedie's formula and measures how well it inpaints noisy inputs – i.e. it measures what the model is designed to do. The energy form additionally requires the normaliser $\log Z_\tau$, a derived quantity that MGD builds (Eq. 21) but never places constraints on, and whose definition is slightly implementation-dependent (bound vs. θ_t-only, see Section 9). The denoiser form has no $\log Z$ dependence at all.

3. *Fewer numerics-side worries.* The energy form takes a numerical derivative $\partial_s\mathbb{E}_y[U|x]$ across the t_{reg} grid; this amplifies Monte-Carlo noise unless common random numbers are used across τ-nodes (Section 11's controlled test showed this was the dominant implementation hazard, and the mitigation changed defaults). The denoiser form has no differentiation step at all. Its one real downside – larger seed-to-seed scatter at $\tau \gg 1$, where d_{eff} itself is small – is cured by seed averaging; a 30-seed production run brings the floor well below the physical mean value at every τ.

Suggested paper phrasing. "*For a sample x and noise scale τ, we report the local effective dimensionality*

$$d_{\text{eff}}(x,\tau) \;=\; \tfrac{1}{\tau}\,\mathbb{E}_\xi\big[\,\|x-\mathbb{E}[x\mid y]\|^2 \;\mid\; x\big]\,, \qquad y = x+\sqrt{\tau}\,\xi,$$

*with the posterior mean given by Tweedie's formula applied to the model's own score, $\nabla_y \log p_\theta(y,\tau) = +\sin\alpha_t\,\theta_t^\top\nabla\varphi(\sin\alpha_t\,y)$. This is Guth, Kadkhodaie & Simoncelli's [*arXiv:2506.05310*] Eq. 11; the equivalent Eq. 12 (an energy-based definition requiring Fokker–Planck for p_θ) differs from Eq. 11 by a residual quantifiable in the moment-matched setting and reported in Appendix X as a diagnostic of model mismatch.*"

The appendix cross-reference, built from Section 10, documents the gap as the integrated FPE residual $R(y,\tau)$ of Eq. 24 and is one short figure showing the gap curve and its concentration at the burst scale. That removes any ambiguity for a reader comparing MGD's d_{eff} to a diffusion-model baseline whose two forms would agree.

13 Dictionary

Guth et al.	MGD (this codebase)
VE noise scale τ	$\tau(t) = \cot^2(\alpha_t)$, $\alpha_t = \pi t/2$
score $\nabla_y \log q_\tau(y)$	$+\sin(\alpha_t)\, \theta_t^\top \nabla\varphi(\sin(\alpha_t) y)$
energy $U(y, \tau)$	$-\theta_t^\top \varphi(\sin(\alpha_t) y) + \log Z_t - d \log \sin(\alpha_t)$
$\log Z_\tau$	$H_{\text{bound}}(t) + \theta_t^\top m_t$ *(bound, Eq. 21)*
noised sample $y = x + \sqrt{\tau}\,\xi$	$\sin(\alpha_t)\, y =$ `compute_interpolant`(x, ξ)
per-τ fit	$\theta_t \equiv$ `Theta_reg[i]`, grid t_{reg}

14 What's used from a saved run

- `Theta_reg`, shape (n_t, r) – the regularised θ_t trajectory (`codes/regularised_theta.py`); used exclusively (not the raw, finer θ_t) for a single self-consistent grid and to avoid a pre-fix off-by-one alignment issue present in some saved runs' raw θ_t/`dH_t_bound` pairing.
- t_{reg} – `Theta_reg`'s own coarse time grid; not saved for the d=256 runs on disk as of this writing, so it is reconstructed offline (`reconstruct_t_reg`) by replaying `SDE.forward_regularised`'s block-accumulation and near-node-pruning exactly, from the saved fine grid t alone – verified to reproduce the on-disk `Theta_reg` length exactly.
- Each run's `fitted_potentials/` – $\varphi, \nabla\varphi$ with the SAME pruned active-statistic set the fit used (some potentials auto-prune during `fit()`; skipping this step silently mis-indexes every θ coefficient).
- `dH_t_bound` + the fine grid t – integrated to $H_{\text{bound}}(t)$ (Eq. 21), needed only by the energy form.
- `aux_moments` (`barphi_e`) if pulled from the cluster, else m_t recomputed from the run's own x_1 – needed only by the energy form.

15 Caveats

- **The energy–denoiser gap is model residual, not an implementation defect** (Section 10). The earlier working hypothesis – that the gap was "bound bias" in $\log Z_t$ – was ruled out on 2026-10-13/14: a Gaussian oracle test with exact inputs (Section 11) reproduces the energy form to the Monte-Carlo noise floor, and on the production runs the gap persists at -50 to -70 (units of d_{eff}, which is dimensionless) across two independent regularisation arms (λ=10^{-3} constrained L_6<0 and λ=5×10^{-4} unconstrained) while the two alternative $\log Z_t$ constructions differ by ≤ 0.3 nats in $\log Z_t$. The gap is the integrated residual $-2\tau\, \mathbb{E}_\xi[R]$ of Eq. 24, concentrated at the burst scale $\tau \sim 10^{-3}$.
- θ_t becomes numerically extreme as $t \to 1$ ($\tau \to 0$): every seed inspected has one dominant, near-null-direction coefficient at the finest scale, with $|\theta|$ up to $\sim 10^8$ on the final runs. This is harmless for the actual energy/score (both stay smooth off-support – checked directly), but $\theta_t^\top \varphi$ and $\theta_t^\top \nabla\varphi$ must be computed in double precision, not the float32 the values are stored in. The current defaults (`denoiser_d_eff(float64=True)`, `energy_d_eff(float64=True)`) do this automatically, with a silent fall-back to float32 for the handful of potentials whose internal buffers do not accept float64 (`Scattering_Fourth_Order_Mod2_*`, memoised per potential

in `_NO_FLOAT64`). Measured $|\text{den32} - \text{den64}|$ on the final runs stays at $\sim 10^{-4}$ while the Monte-Carlo noise is ~ 1 – precision is not currently a bottleneck.

- The finest available τ is set by how close to $t = 1$ the run's schedule and `n_subsample` block reach, not by an arbitrary choice – see `codes/effective_dimension.py`'s `reconstruct_t_reg` for the exact grid actually available for a given run.

Implementation changelog

- *2026-09-23:* `denoiser_d_eff`/`energy_d_eff` cast θ_t to `x.dtype` for `potentials.grad()` but not to `x.device` – harmless on a single-device (CPU) run, but a hard crash on Jean Zay, where θ_t/m_t load onto CPU (`map_location='cpu'`) while `x`/the potentials live on `cuda`. Fixed: both are now moved to `x.device` inside each function.

- *2026-09-24:* the differentiation bug and fix described in Section 11 (`local_poly_derivative` replacing a raw `numpy.gradient` call in `energy_d_eff`).

- *2026-10-13:* `energy_d_eff` and `denoiser_d_eff` default to `float64=True`; the energy form additionally defaults to `common_noise=True` (same Gaussian noise draw ξ at every τ-node, so the pointwise $\partial_s \mathbb{E}_y[U|x]$ differentiation does not amplify Monte-Carlo noise). A helper `_call_maybe_f64` inside `effective_dimension.py` remembers per-potential which basis classes refuse float64 input (notably `Scattering_Fourth_Order_Mod2_Real/Imag_Q1`, whose internal buffers are float32 only), and falls back to float32 for those while keeping the rest in float64. New helpers `phi_stack` and `h_from_theta_reg` added; the latter is the θ_t-only alternative $\log Z_t$ construction discussed in Section 9.

- *2026-10-14:* the earlier working hypothesis that the energy–denoiser gap was "bound bias" in $\log Z_t$ (previous version of Section 5) was ruled out and replaced with the FPE-residual explanation of Section 10: the gap persists at the same -50 to -70 level (units of d_{eff}) across two independent regularisation arms with different λ and different L_6 sign-constraint choices, while the two possible $\log Z_t$ constructions differ by ≤ 0.3 nats in $\log Z_t$. The notes' recommendation to use the denoiser form as the primary estimator (Section 12) is a consequence of this. Diagnostic script: `deff_diagnostics/deff_latest_run_check.py` (per-seed CSVs); CSV-loader notebook: `turbulence/effective_dimension_deff_final.ipynb`.

- *2026-10-15:* Part I added. Eq. 24 corrected. The previous version had the sign of the score-magnitude term inconsistent with the other two terms. It also wrote the family in y instead of $z = \sin \alpha_t y$, so it lacked the rescaling term (iv) and the $dt/d\tau$ factor. The new version is checked to vanish identically in the Gaussian case. The gap is expressed in units of d_{eff} (dimensionless), not nats. The stale "lower-bound bias" wording in the `energy_d_eff` docstring is replaced. Later the same day, Part I was rewritten in MGD variables only (no auxiliary y, z, s). The new Section 4 lists the subtleties of the denoiser-to-MGD translation. Among them: θ_k is a score-matching estimate of the walker law, which refines Part II's "moment-matched, not score-matched".